

Virtualization & Cloud (Part-1)

- Topics covered:
 - Intro To Virtualization
 - Virtual Servers
 - Virtual Networks
 - Intro To Cloud Computing
 - Essential Characteristics
 - Service Models
 - Deployment Models
 - Connecting To Public Clouds

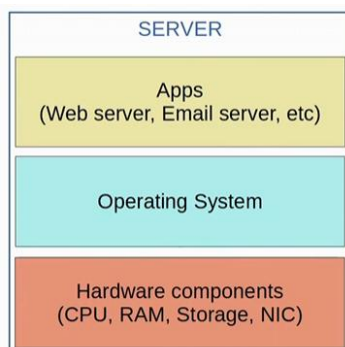
Intro To Virtualization

Server Hardware and Traditional Servers

Although Cisco is best known for networking devices such as routers, switches, and firewalls, it also offers hardware servers through its UCS (Unified Computing System) product line. Other major server vendors include Dell EMC, HPE (Hewlett Packard Enterprise), and IBM. Data centers typically contain racks filled with servers from these and other vendors.

Before virtualization, there was a one-to-one relationship between a physical server and an operating system.

Traditional Server Architecture



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A single operating system runs directly on the server hardware, and applications such as web servers and email servers run on top of that operating system. Traditionally, separate physical servers were often used for separate services such as web, email, and database servers.

Running all services on a single operating system is possible, but it is not ideal because a problem in one application could affect all other applications. However, dedicating one physical server to each application is inefficient because:

- Physical servers are expensive.
- They consume space and power.
- Resources such as CPU, RAM, storage, and NIC capacity are often underutilized.

A NIC (Network Interface Card) is the component that connects a server to the network.

Virtualization

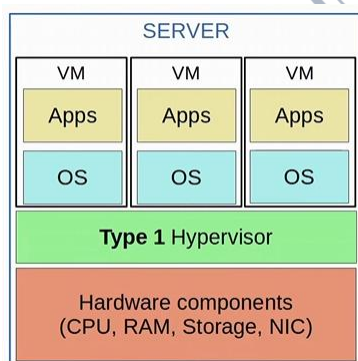
Virtualization breaks the one-to-one relationship between hardware and operating systems by allowing multiple operating systems to run on a single physical server. Each operating system instance is called a Virtual Machine (VM).

Type 1 Hypervisor

A hypervisor manages and allocates hardware resources such as CPU and RAM to virtual machines. Another name for a hypervisor is VMM (Virtual Machine Monitor).

A Type 1 hypervisor runs directly on top of the hardware.

Type 1 Hypervisor Architecture



Examples: VMware ESXi, Microsoft Hyper-V

Type 1 hypervisors are also called:

- Bare-metal hypervisors
- Native hypervisors

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They are commonly used in data centers because they run directly on hardware and require very few resources themselves, leaving more resources available for virtual machines.

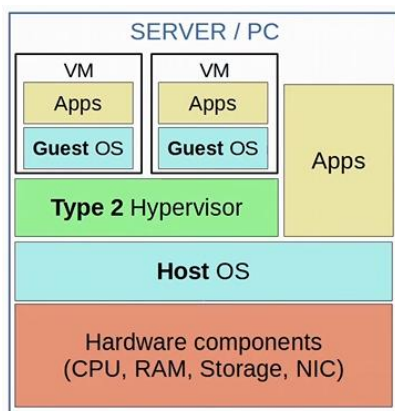
Type 2 Hypervisor

Type 2 hypervisors run as regular programs on an operating system.

Examples:

- VMware Workstation
- Oracle VirtualBox

Type 2 Hypervisor Architecture



The operating system running directly on the hardware is called the **Host OS**, while the operating system running inside a VM is called the **Guest OS**.

Type 2 hypervisors are also called **Hosted Hypervisors**. They are rarely used in data centers but are common on personal computers.

Benefits of Virtualization

According to VMware, virtualization provides the following benefits:

- 1. Partitioning:** Multiple operating systems can run on one physical machine while sharing system resources. This helps solve hardware underutilization.
- 2. Isolation:** Virtual machines are isolated from one another. If a fault occurs in one VM, it does not affect other VMs.
- 3. Encapsulation:** A virtual machine can be saved, copied, and moved easily, similar to a regular file.
- 4. Hardware Independence:** Virtual machines can be created or moved to any physical server that supports the required hypervisor.

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Additional Advantages

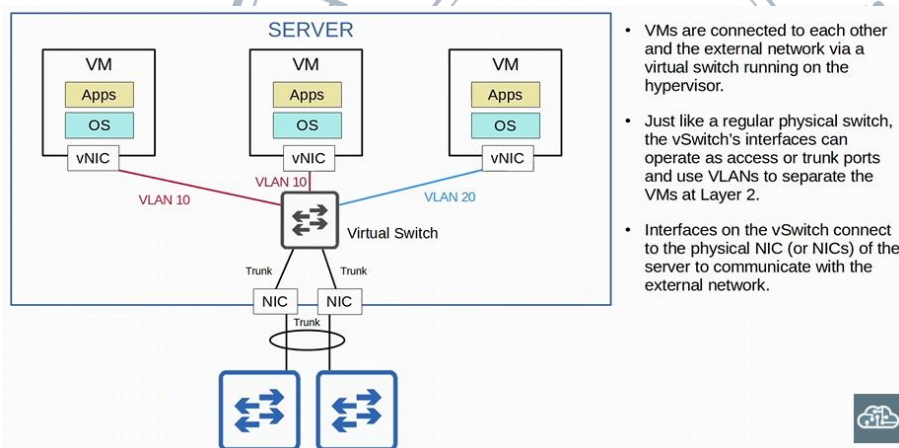
Virtualization provides:

1. Reduced capital costs because fewer physical servers are needed.
2. Reduced operating costs through lower space, power, and cooling requirements.
3. Faster deployment of new systems.
4. Reduced or eliminated downtime through redundancy.
5. Increased efficiency, agility, and responsiveness.
6. Faster provisioning compared to purchasing and deploying new physical servers.

Virtual Networks

Virtual machines connect to each other and to external networks through a virtual switch (vSwitch) running on the hypervisor.

Virtual Network Architecture



Key points:

1. VMs connect through a virtual switch running on the hypervisor.
2. A hypervisor may provide its own virtual switch, or a Cisco virtual switch can be used.
3. Virtual switch interfaces can operate as access ports or trunk ports.
4. VLANs can be used to separate VMs at Layer 2.
5. In the example, two VMs belong to VLAN 10 and one VM belongs to VLAN 20.
6. The virtual switch connects to one or more physical NICs to reach external networks.
7. A vPC (Virtual Port Channel) can connect the server to two physical switches for redundancy. This is commonly used in data centers.

Intro To Cloud Computing

Traditional Infrastructure Deployment Models

Before cloud computing became common, organizations generally used one or both of the following approaches.

1. On-Premises

All infrastructure is located on company property.

This includes:

- Servers
- Network devices
- Routers
- Firewalls
- Web servers
- Database servers

The company owns the equipment and is responsible for providing space, power, and cooling.

2. Colocation

A colocation data center rents space to customers for placing their infrastructure.

The data center provides:

- Space
- Electricity
- Cooling
- Physical security

However, the customer's servers and network devices remain the customer's responsibility.

Traditionally, companies used either on-premises infrastructure or a combination of on-premises and colocation facilities. Cloud services later emerged as a popular alternative.

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Cloud Computing

According to NIST (National Institute of Standards and Technology), cloud computing is defined in Special Publication 800-145. The cloud model consists of:

1. Five essential characteristics
2. Three service models
3. Four deployment models

For CCNA, these are important cloud concepts. A basic understanding of these topics is sufficient; knowledge of specific services from individual cloud providers is not required.

Five Essential Characteristics of Cloud Computing

A service is generally not considered a true cloud service unless it has all five of these characteristics:

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

Let's briefly look at each one.

1. On-Demand Self-Service

NIST Definition

A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.

The customer can start or stop using services through a web portal without directly interacting with the service provider.

For example, when setting up virtual servers on AWS, the customer logs into the web portal and performs the setup independently, without requesting quotes, arranging payments manually, or waiting for provider staff to configure the servers.

2. Broad Network Access

NIST Definition

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Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).

Cloud services are available through standard network connections such as:

- Internet
- Private WAN connections

They can be accessed from many device types, including PCs, smartphones, tablets, laptops, and workstations.

Cloud-hosted resources, such as virtual servers, can also be reached through the Internet or various WAN connectivity options.

3. Resource Pooling

NIST Definition

The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.

Examples of pooled resources include:

- Storage
- Processing
- Memory
- Network bandwidth

The provider maintains a large shared pool of resources. When a customer requests a service, such as creating a VM, the required resources are allocated from that shared pool.

Customers generally do not know the exact location of the resources being used, although they may be able to specify a higher-level location such as a country, state, or data center.

Summary: A shared resource pool is maintained by the cloud provider, and customer requests are fulfilled by allocating resources from that pool. When a VM is created, a portion of the shared resources is assigned to that VM.

4. Rapid Elasticity

NIST Definition

Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand.

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Customers can quickly:

- Expand services
- Add new VMs
- Increase storage
- Reduce resources when no longer needed

The available resource pool should appear virtually unlimited from the customer's perspective, allowing resources to be obtained in any quantity whenever required.

In addition to scaling up quickly, customers should also be able to scale down quickly when demand decreases.

5. Measured Service

NIST Definition

Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service.

Measured resources may include:

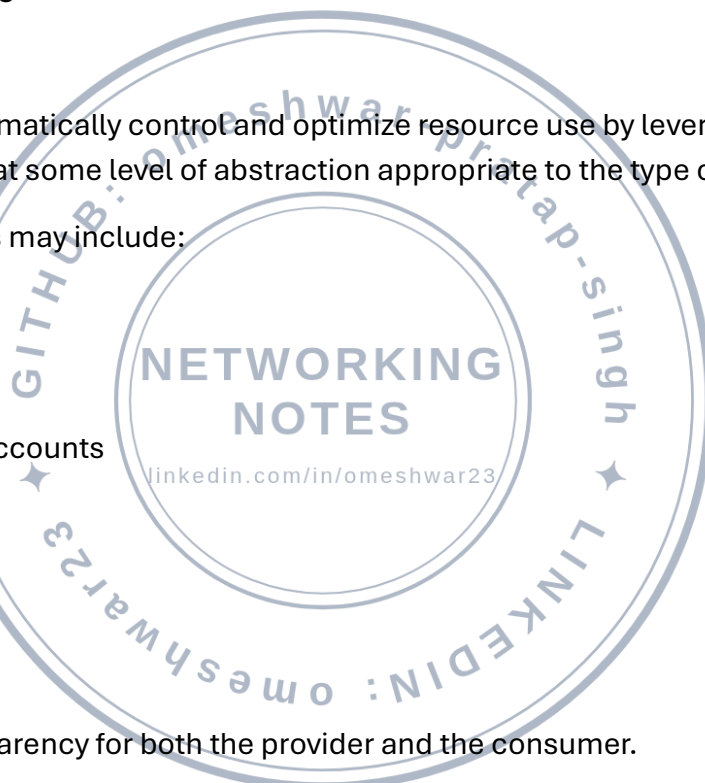
- Storage
- Processing
- Bandwidth
- Active user accounts

Resource usage is:

- Monitored
- Controlled
- Reported

This provides transparency for both the provider and the consumer.

Customers can view their usage through the web portal and are charged according to actual usage. Usage is clearly visible so customers can monitor their resource consumption.



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Three Cloud Service Models

Cloud computing delivers resources through service models. Instead of purchasing physical infrastructure and managing everything directly, the provider offers these resources as services.

The three primary cloud service models defined by NIST are:

1. Software as a Service (SaaS)
2. Platform as a Service (PaaS)
3. Infrastructure as a Service (IaaS)

Software as a Service (SaaS)

NIST Definition

The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure.

Applications are accessed through:

- Web browsers
- Program interfaces

The consumer does not manage or control:

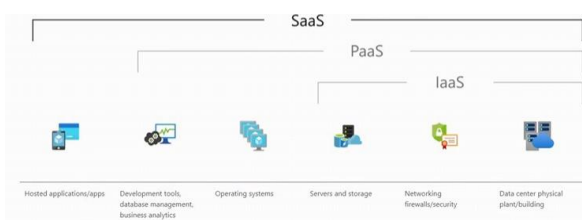
- Network infrastructure
- Servers
- Operating systems
- Storage
- Application capabilities

Except for limited user-specific application settings.

Examples mentioned:

- Microsoft Office 365
- Google G Suite (including Gmail)

SaaS Responsibility Model



When using SaaS, the provider manages everything from the physical data center to the applications themselves. The customer simply uses the application being provided.

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Platform as a Service (PaaS)

NIST Definition

The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider.

The consumer does not manage or control:

- Networks
- Servers
- Operating systems
- Storage

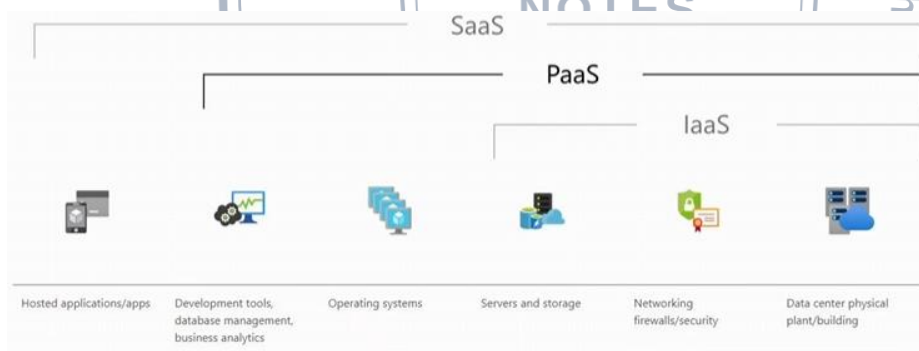
However, the consumer does control:

- Deployed applications
- Some application-hosting configuration settings

PaaS provides a platform that developers can use to create applications.

Examples mentioned: AWS Lambda, Google App Engine

PaaS Responsibility Model



Infrastructure as a Service (IaaS)

NIST Definition

The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications.

The consumer:

- Can deploy and run operating systems and applications.
- Has control over operating systems, storage, and deployed applications.

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- May have limited control of some networking components (such as host firewalls).

The consumer does **not** manage or control the underlying cloud infrastructure.

Examples of IaaS services include: Amazon EC2, Google Compute Engine

Creating virtual machines on AWS is an example of Infrastructure as a Service.

The service provider manages:

- Physical data center
- Networking and security
- Server infrastructure
- Storage infrastructure

This provides infrastructure that customers can use for almost any purpose. IaaS provides the highest level of customer control among the three cloud service models.

Cloud Service Models Comparison



The three fundamental cloud service models are:

1. Software as a Service (SaaS)
2. Platform as a Service (PaaS)
3. Infrastructure as a Service (IaaS)

Many other "*something as a service*" offerings exist today, but these three form the foundation of cloud services.

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Four Cloud Deployment Models

Many people associate cloud computing only with public cloud providers such as AWS, Azure, and GCP. However, public cloud is only one deployment model.

The four deployment models are:

1. Private Cloud
2. Community Cloud
3. Public Cloud
4. Hybrid Cloud

1. Private Cloud

NIST Definition

The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers.

A private cloud:

- Is used exclusively by a single organization.
- May be owned, managed, or operated by:
 - The organization itself
 - A third party
 - A combination of both
- May exist:
 - On-premises
 - Off-premises

Private clouds are generally used by:

1. Large enterprises
2. Government organizations

Although the infrastructure may belong to a third party, it is reserved exclusively for one organization.

Private clouds provide the same cloud service models:

- SaaS
- PaaS
- IaaS

However, the cloud infrastructure is dedicated to a single organization.

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2. Community Cloud

NIST Definition

The cloud infrastructure is provisioned for exclusive use by a specific community of consumers from organizations that have shared concerns.

These shared concerns may include:

- Mission
- Security requirements
- Policies
- Compliance considerations

Characteristics:

- Similar to private cloud.
- Reserved for a specific group of organizations.
- May be owned, managed, or operated by:
 - One or more organizations in the community
 - A third party
 - A combination of both
- May exist on-premises or off-premises.

This is the least common cloud deployment model.

3. Public Cloud

NIST Definition

The cloud infrastructure is provisioned for open use by the general public.

Characteristics:

- Available for public use.
- Exists on the premises of the cloud provider.
- May be owned, managed, or operated by:
 - Businesses
 - Academic organizations
 - Government organizations
 - Combinations of these

This is the most common cloud deployment model.

Popular public cloud providers mentioned: AWS, Azure, GCP, OCI, IBM, Alibaba

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AWS has been the dominant provider for a long time.

4. Hybrid Cloud

NIST Definition

The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities but are bound together.

Hybrid cloud:

- Combines two or more deployment models.
- Is not a completely separate type of cloud.
- Enables data and application portability between cloud environments.

Example mentioned: A private cloud that offloads workloads to a public cloud when additional resources are needed.

Benefits of Cloud Computing

1. Cost

Cloud computing reduces or eliminates:

- Hardware purchases
- Software purchases
- Data center setup costs

Instead of large capital expenses (CapEx), costs become operational expenses (OpEx), consisting of smaller ongoing payments.

For many organizations, this can reduce overall costs.

2. Global Scalability

Cloud services can scale rapidly across geographic regions.

When creating services such as virtual servers, customers can choose:

- Country
- Region within the country

This can improve service availability for users in those locations.

3. Speed and Agility

Cloud services are:

- Available on demand
- Rapidly provisioned

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Large amounts of resources can be deployed within minutes.

4. Improved Productivity

Cloud services eliminate many time-consuming tasks such as:

- Procuring physical servers
- Installing servers in racks
- Cabling
- Installing operating systems
- Updating operating systems

This can also contribute to lower costs.

5. Reliability

Cloud services simplify:

- Backups
- Disaster recovery

Data can be mirrored across multiple geographic locations.

If one server fails, copies of the data remain available elsewhere.

The cloud is not always the best solution for every organization. Many companies use a combination of:

- On-premises infrastructure
- Colocation facilities
- Public cloud services

Cloud benefits should be evaluated alongside potential drawbacks before making decisions.

Connecting To Public Clouds

Connecting to Public Cloud Resources

Enterprises can connect to public cloud resources using multiple WAN options.

Examples include:

- Private WAN connections
- Internet connections

A private WAN example is: MPLS VPN

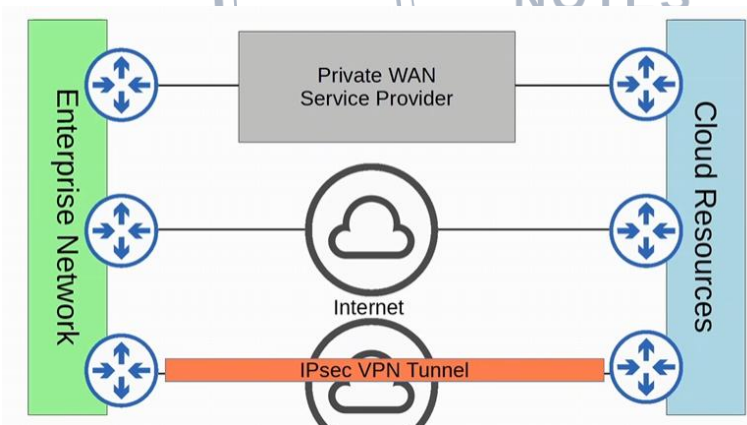
Connecting through the Internet is:

- Cheap
- Flexible
- Easy

However, it is less secure than a private WAN connection.

Secure Internet Connectivity

A VPN can provide a secure connection over the Internet to cloud resources.



Key points:

1. Cloud resources can be reached through a private WAN.
2. They can also be reached through the Internet.
3. VPNs provide secure communication over the Internet.
4. Redundant connections are preferred.
5. Single points of failure should be avoided whenever possible.